

Inter molecular hydrogen bonding

In fact, most of the carboxylic acids exist as dimer in its vapour phase.

iii) Lower aliphatic carboxylic acids (up to four carbon) are miscible with water due to the formation of hydrogen bonds with water. Higher carboxylic acid are insoluble in water due to increased hydrophobic interaction of hydrocarbon part. The simplest aromatic carboxylic acid, benzoic acid is insoluble in water.

iv) Vinegar is 6 to 8% solution of acetic acid in water. Pure acetic acid is called **glacial acetic acid.** Because it forms ice like crystal when cooled. When aqueous acetic acid is cooled at 289.5 K, acetic acid solidifies and forms ice like crystals, where as water remains in liquid state and removed by filtration. This process is repeated to obtain glacial acetic acid.

12.12 Chemical properties of carboxylic acids.

Carboxylic acid do not give the characteristic reaction of carbonyl group $\diagdown C = O$ as given by the aldehydes and ketones. as the carbonyl group of carboxylic acid is involved in resonance:

The reactions of carboxylic acids can be classified as follows:

A) Reactions involving cleavage of O – H bond.

B) Reactions involving cleavage of C – OH bond.

C) Reactions involving – COOH group.

D) Substitution reactions involving hydrocarbon part.

A) Reactions involving cleavage of O – H bond.

1) Reactions with metals

Carboxylic acid react with active metals like Na, Mg, Zn etc to form corresponding salts with the liberation of hydrogen.

Example



Acetic acid — Sodium acetate

2) Reaction with alkalies

Carboxylic acid reacts with alkalies to neutralise them and form salts.

XII U12-Carbonyl Compounds.indd 172 08-12-2021 22:50:43